

EDUCATION

- **International Institute of Information Technology, Hyderabad** Hyderabad, India
B.Tech + M.S. (by research) in Computer Science and Engineering *September 2022 – May 2027*
CGPA: 8.19
 - Undergraduate Researcher at **Precog Research Lab**
 - Coordinator, **Programming Club**
 - Teaching Assistant for **Data Structures and Algorithms** Course Spring'25

EXPERIENCE

- **Google** Bangalore, India
Software Engineering Intern *May - July 2026*
 - Modernized a critical datacenter modeling service by migrating its monolithic structure to an asynchronous microservices architecture, implementing non blocking database queries to eliminate cascading latency issues.
- **Luddy Center for Artificial Intelligence, Indiana University** Bloomington, Indiana
Visiting Scholar *January - April 2026*
 - Working towards creating stable belief LLMs to enable reliable and consistent human behavior simulation by injecting non factual beliefs in LLMs
- **Microsoft Research** Bangalore, India
Research Intern *July - September 2024*
 - Developed Labeling Copilot – a unified framework for rapid synthetic data creation and annotation, integrating data discovery, augmentation, and foundation models to overcome data scarcity and enable the development of robust computer vision models across industries.
- **SmartTerra** Remote
Machine Learning Intern *January - May 2024*
 - Developed a conversational chatbot to assist non-technical users in analyzing MongoDB datasets via natural language queries. Enabled in-depth data visualization and manipulation with large datasets, without requiring manual scripting or data analysis expertise.
- **Indian Computing Olympiad** Remote
Problem Setter and Author *2023*
 - Created problems for Zonal Informatics Olympiad'2023 and published articles for the official INOI (Indian National Olympiad in Informatics) guide including topics like basic graph theory

PUBLICATIONS

- [3] **SEMMA: A Semantic Aware Knowledge Graph Foundation Model** [PDF]
Arvinth Arun*, **Sumit Kumar***, Mojtaba Nayyeri, Bo Xiong, P. Kumaraguru, Antonio Vergari, Steffen Staab
2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)
- [2] **Labeling Copilot: Tooling for a Rapid Synthetic Data Creation & Annotation** [PDF]
Debargha Ganguly*, **Sumit Kumar***, Ishwar Balallapanawar*, Shashank Kambhatla, Srinivasan Iyengar, Shivkumar Kalyanaraman, Vipin Chaudhary, P. Kumaraguru
2025 IEEE International Conference on Big Data
- [1] **A Cognac shot to forget bad memories: Corrective Unlearning in GNNs** [PDF]
Varshita Kolipaka*, Akshit Sinha*, Debangan Mishra, **Sumit Kumar**, Shashwat Goel, Arvinth Arun, P. Kumaraguru
42nd International Conference on Machine Learning (ICML 2025)

SKILLS SUMMARY

- **Languages:** C, C++, Java, Python, Kotlin, Assembly, HTML, CSS, Javascript, Flask, SQL, Bash, Multithreading
- **Skills:** Competitive Programming, ML, Web Development, Shell scripting, DBMS, Git, PyTorch, MongoDB, Linux, OOPS

ACHIEVEMENTS

- **International Collegiate Programming Contest (ICPC):** Secured a national rank of 28 out of 2500 Indian college teams, qualified for Asia Amritapuri Regionals 2022, awarded with honorable mention
- **Indian National Olympiad in Informatics(INOI):** Ranked top 30 nationally, awarded a Silver medal at INOI 2022, prepared 4+ years for IOI
- **Football:** Awarded Best Player in the college for the year 2024 and I play as center attacking midfield for the college football team.
- **Google Code-In 2019:** Collaborated with various open source organisations and completed 18+ tasks
- **Meta HackerCup 2022:** Qualified for Round 2, secured rank 1274 out of 30,000 participants
- **Codechef Snackdown 2022:** Qualified for Round 2 of Global Programming Contest conducted by Codechef
- **Max rating of 1894(Expert) on Codeforces.:** Top 4% globally
- **Max rating of 2023(5*) on Codechef.:** Top 0.9% globally
- **CodeChamps - Codingal:** Secured rank 1 out of 187 students globally